

Evidence-First Trade-Based Money Laundering (TBML) Triage System

An evidence-led way to decide which unusual public trade patterns to review first

This analytical prototype turns official-derived international trade statistics into a ranked queue of unusual exporter–importer–product–year patterns for human review, with recomputable evidence and an explicit caveat attached to every ranked record.

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Date	27 July 2026
Analytical scope	3 HS6 product families · 2017–2024 · 25,844 annual trade observations
Deployed application	https://tbml-review-triage.streamlit.app/
GitHub repository	https://github.com/jessybaolin/evidence-based-tbml-corridor-drift

HUMAN-REVIEW BOUNDARY

This output prioritises an unusual corridor-product pattern for human review. It does not establish money laundering, misinvoicing, or criminal intent.

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What this project is

An analytical prototype that reads **public, official-derived international trade statistics** and returns a **ranked queue of unusual exporter–importer–product–year patterns for human review**, with recomputable evidence attached to every ranked record.

It is not a detection system, and it does not decide that anything is suspicious. It decides what an analyst should look at first, and it shows its working.

25,844 ANNUAL TRADE OBSERVATIONS 2017–2024, across 4,952 corridors and 3 HS6 product families	Top 50 REVIEW QUEUE The ranked records an analyst opens first, filterable and exportable	4 per case RECOMPUTABLE EVIDENCE ROWS 200 in total — each a metric, a comparison, a threshold and a caveat
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The decision problem is not a shortage of things that look unusual. It is deciding which of them deserve scarce review time, and giving the analyst enough prepared evidence to begin a meaningful review rather than a reconstruction. Anti-financial-crime teams already hold the records that decide a case; what public trade data adds is an independent outside view of the corridor, the product and the period, with the reasons for the ranking stated in advance.

Of the 25,844 observations in scope, **23,656 (91.5%)** carry the positive declared value and quantity an implied-unit-value comparison requires. The rest are retained in full for audit, never scored, and never treated as suspicious for being incomplete.

What it does not conclude. The data is annual, country-to-country, product-level aggregate: no invoices, no shipments, no customers, no payments, no beneficial ownership, no confirmed money-laundering outcomes. A high rank means one thing — this pattern is unusual on measurable public signals, and here is exactly why. Everything after that requires bank-held records and human judgement.

SECTION 01

Why the problem matters

The hard part is not finding one unusual number. It is deciding what deserves scarce review time, and being able to say why.

Trade-based money laundering is the movement or concealment of illicit value through trade — including manipulation of price, quantity or the description of goods. It is difficult to assess from any single vantage point. Volumes are enormous, legitimate prices move with grade, contract terms, freight and timing, and no one party sees the whole physical and financial chain at once. A bank sees its customer's documents and payments. Public statistics see the country-to-country flow. Neither sees the other.

That gap is where an independent outside signal is useful. Public trade statistics can say that a corridor moved differently from its own history, from its same-product peers, and from the wider market in the same year — an observation a bank cannot make from its own book alone.

Why an unexplained score creates work rather than saving it

An alert with a number attached transfers the work to the analyst. Before reviewing anything, they have to reconstruct what changed, what the record was compared against, whether the whole market moved that year, and whether weak data explains the result. That reconstruction is slow, it is done differently by different reviewers, and it is hard to defend afterwards.

This project addresses that first mile. Every ranked record arrives with the facts that drove the ranking, the comparison each was measured against, the threshold it passed, the data-quality status of the row, and a plain statement of what public data cannot answer. The opening question changes from *“why did the system produce this?”* to *“which stated fact should I check first?”*

THE REVIEW CHALLENGE

- Far more public trade flows than can be inspected by hand
- A market-wide price move can look like a corridor-specific one
- Aggregate data can be incomplete, or commercially ambiguous, or both
- An unexplained score has to be reconstructed before it can be used

WHAT THE PROJECT DOES ABOUT IT

- Ranks the strongest unusual-pattern signals so capacity goes to the tail
- Separates a corridor's own movement from the market's
- Reports data quality beside review priority, never merged into it
- Attaches the comparison, the threshold and the caveat to every ranked record

Why this is a wholesale-banking problem

Trade finance and correspondent banking are where a corridor, a commodity and a counterparty meet a payment. Those teams already combine commercial documents, customer knowledge and payment information; what they lack is an independent reason to ask a sharper question about a particular corridor, product or period before the deeper review begins. That is the specific contribution here — not a replacement for internal controls, and not a second opinion on a customer, but an outside prompt with its reasoning attached.

KEY POINT

The value being offered is disciplined triage: a defensible order of work, a consistent starting structure for every review, and an auditable link from a ranking back to the public data that produced it.

SECTION 02

Data foundation and product scope

Trade facts, market context and review guidance are kept apart on purpose, so that guidance or a benchmark difference can never be mistaken for evidence about a record.

CEPII BACI OFFICIAL-DERIVED TRADE FACTS	World Bank Commodity Markets Outlook COMMODITY MARKET CONTEXT	FATF–Egmont TBML publications TYPOLOGY AND REVIEW CONTEXT
Annual bilateral trade value and quantity by exporter, importer and HS6 product, built from national statistics reported through UN Comtrade. This is the only source that contributes rows to the analysis. Boundary. Annual public aggregates, never transactions.	Annual benchmark prices for palm oil, copper and gold, used to ask whether a corridor moved differently from the wider market in the same year. Boundary. Broad market context, never invoice-level fair value.	Published guidance that shapes caveats, possible benign explanations and the questions a reviewer should ask next. Boundary. Never a model input, a label, or a row-level fact.

A fourth input carries no analytical weight of its own: BACI's country codes and the HS17 product catalogue resolve numeric codes to names. *HS6* is the six-digit Harmonized System code that identifies a traded product the same way across countries, and the project uses one HS6 code per family so a comparison always holds the product constant.

The scope boundary matters more than the scale. There are no customer records, no invoices, no individual payments, no shipment-level records, no beneficial-ownership information and no confirmed TBML labels anywhere in the data. Nothing here can establish that something occurred — only that a measured public pattern is unusual against a stated reference.

Why these three commodities

Crude palm oil HS6 151110	Refined copper cathodes HS6 740311	Non-monetary unwrought gold HS6 710812
A high-volume regional bulk commodity, concentrated in Southeast Asian routes. It tests whether the ranking stays measured around very large, established, ordinary flows. Boundary. Volume is not a risk signal.	A standardised industrial metal with a widely followed and broadly interpretable benchmark. It gives the cleanest test of market-relative comparison. Boundary. A standard grade still varies by contract.	Compact, high value per tonne, prominent in published typologies, and the family with the most reporting-coverage difficulty. It stress-tests pronounced implied-price patterns. Boundary. Prominence in a typology is not evidence.

Together they test different trade profiles rather than treating one commodity as a stand-in for all trade. Gold carries **81.0%** of the declared value in scope, copper **16.3%** and palm oil **2.7%** — three very different shapes of market, deliberately.

CAUTION

Selecting a product family, a country or a corridor for study does not make it suspicious. These three were chosen because they differ from one another in ways that test the method, and because each has a published benchmark the comparison can rest on.

SECTION 03

How the solution works

The unit of analysis is deliberately small and completely traceable: one exporter, one importer, one product, one year.

Every figure in the project rests on one kind of row — a single **corridor-product-year**: one exporter country, one importer country, one HS6 product and one calendar year, with the declared value and quantity aggregated across that public annual flow. Dividing value by quantity gives an *implied unit value*: an average derived from an aggregate, not a price observed on any invoice. Each row is prepared once and carries its matching same-year commodity benchmark from the start, so the reference used later for comparison is fixed before any comparison is made.

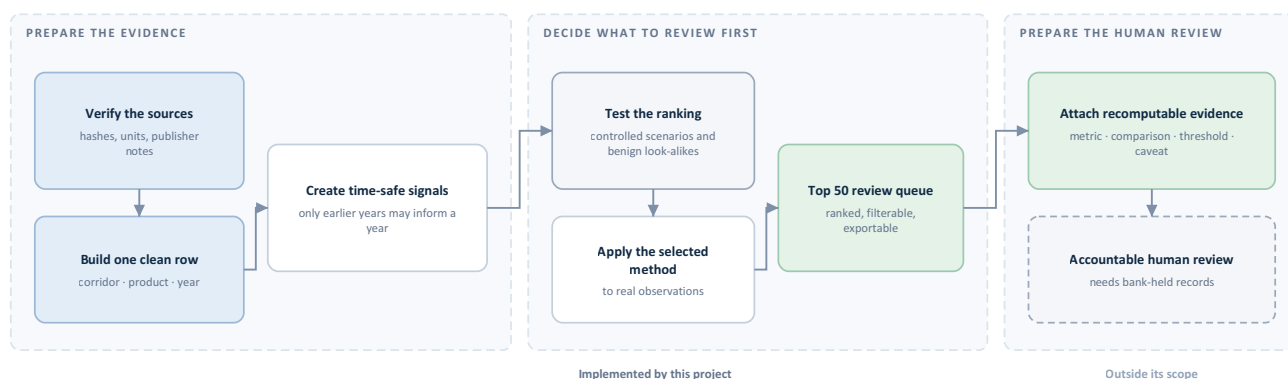


Figure 1. How the project turns public trade statistics into a prepared review. The three bands are the three jobs: prepare the evidence, decide what to review first, and hand over to a person. Only the first two are implemented here — the last requires records this project does not have and is drawn dashed throughout.

Two steps in that flow carry most of the discipline. **Time safety**: a record for a given year may only be informed by that year and earlier, so historical baselines are shifted and a row is never assessed using its own future. **Separation of evaluation from production**: the scenarios used to test the ranking live in a separate copy of the frozen panel, and no scenario row and no scenario label ever reaches the feature table, the review queue, the evidence table or the dashboard.

What the ranking is, and what it is not

Several independent comparisons are combined into a single review order. Transparent weighted rules were compared against learned methods under a strict time discipline — earlier years to fit, later years held back — and the retained approach blends the two, with the blend chosen before the final test years were opened.

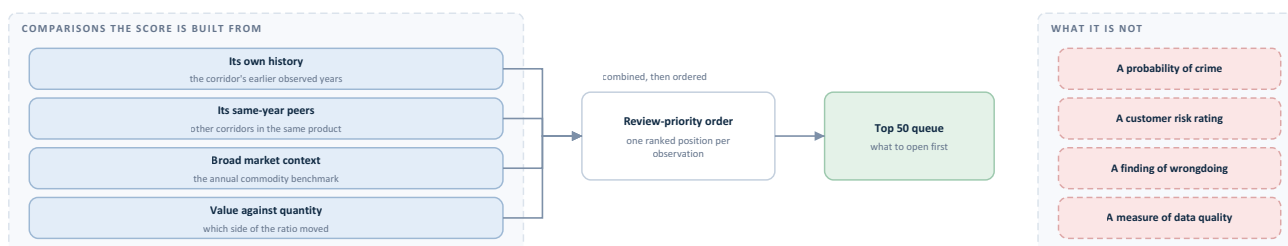


Figure 2. What a review-priority position is built from, and the four things it is deliberately not. Each comparison on the left is recomputable from the published data; the combined result orders the population, and the top 50 are published as the queue.

On the held-out 2023–2024 test years the selected method placed **24 of the 50 highest-ranked rows** on injected review-priority scenarios, against about **0.90** expected from reviewing 50 at random — a concentration factor of **26.8×** — and no deliberately benign look-alike entered the top 50. Those are results on a controlled synthetic evaluation: they measure whether the ranking finds patterns that were planted, and they are not detection rates.

Records without a usable positive quantity cannot support an implied-unit-value comparison at all. They are excluded from scoring, retained for audit, and reported on their own dashboard page — so a coverage gap is never quietly converted into an anomaly.

CAUTION

The review-priority score is a ranking signal. It is not a calibrated probability of money laundering, not a customer risk rating, and not a finding of wrongdoing. A higher position means *review this earlier*, and nothing more.

What travels with a ranked record

A position on its own would be exactly the unexplained score this project set out to avoid. Every queued record therefore carries **4 recomputable evidence rows** — 200 across the queue — and each row is built the same way: the observed metric, the reference it was compared against, the threshold it passed, a severity, a plain-English summary, and a caveat naming what could otherwise explain the value. A reviewer opening any case finds the same six things in the same order, which is what makes one case comparable with the next and what lets a reviewer disagree with the reasoning rather than only accept or ignore it.

SECTION 04

The dashboard journey

The interface narrows deliberately — scope, then market context, then a ranked queue, then a single case — and it keeps validation and coverage on separate, optional tracks.

The deployed application is a 10-page read-only presentation layer over the pipeline’s generated files, plus a welcome route. It never scores anything, never retrains anything and never writes back into the analysis — changing a filter re-slices a cached table. Every page carries the human-review boundary as a fixed ribbon, and each product family is read on its own scale so chart heights are never compared across panels.

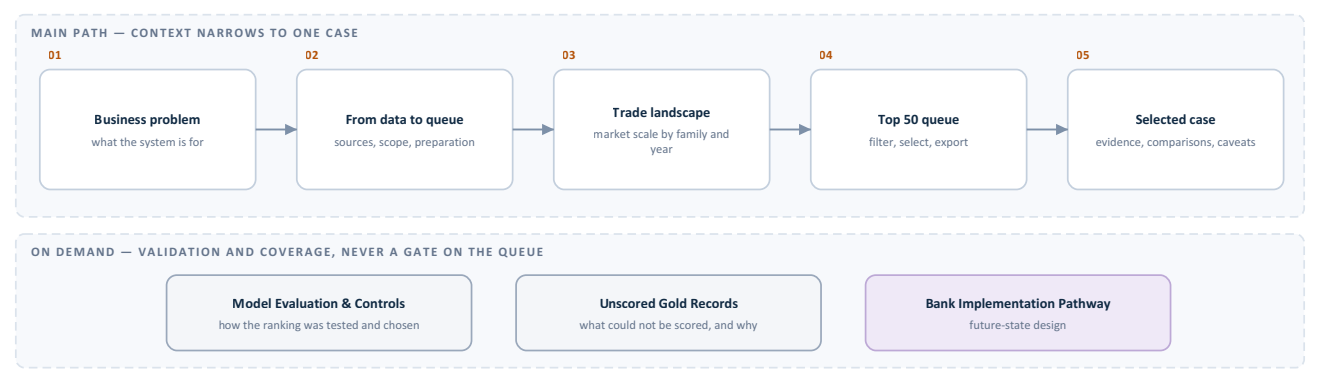


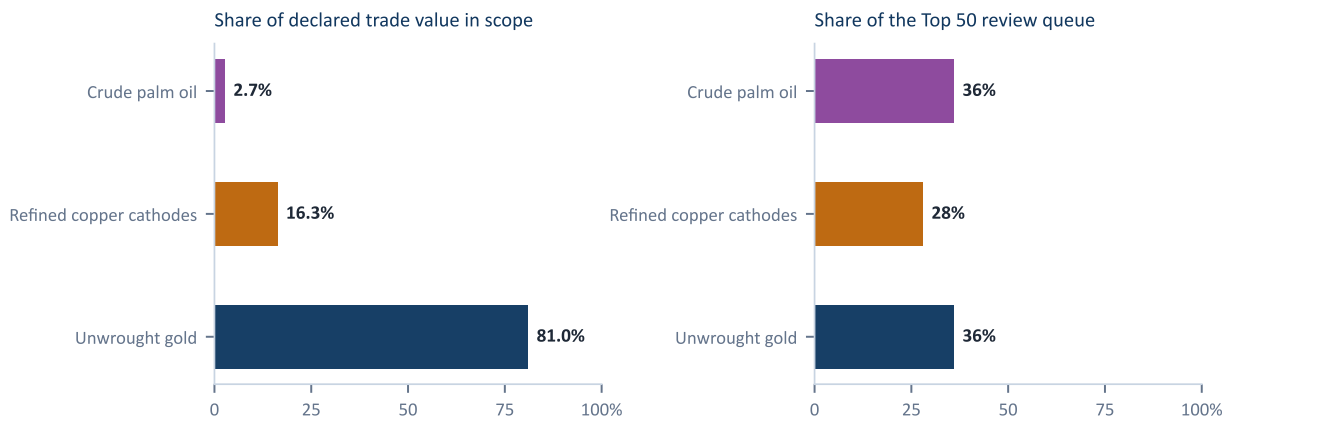
Figure 3. The stakeholder journey through the current application, with the page names as they appear in the product. Model Evaluation & Controls and Unscored Gold Records are on-demand tracks: they are available at any point and are never a gate the reader must pass before reaching the queue.

Four populations, deliberately not the same number

The most common misreading of a queue is to treat it as a share of the market. It is not, and the application separates the four populations on purpose.

Table 1. The four populations a reader moves through, and what each one is.

Population	Size	What it is
Observations in scope	25,844	Every annual corridor—product record across 2017–2024, carrying \$3.52 trillion of declared value
Scored population	23,656	Those with the positive value and quantity a unit-value comparison needs (91.5% of scope)
Top 50 review queue	50	The highest-ranked records, published with evidence — what an analyst opens first
Selected case	1	One record examined in depth, with its comparisons, evidence and caveats



Left: share of the 3.52 trillion US dollars of declared value in the eight-year scope. Right: share of the 50 queued records (crude palm oil 18 · refined copper cathodes 14 · unwrought gold 18). Shares of two different totals, shown side by side; the bars are not a ratio of one another.

Figure 4. Market scale is not review priority. Gold carries most of the declared value in scope but takes a share of the queue close to the other two families, because the ranking measures how a record behaves against its own references — not how large it is.

What the queue page gives a reviewer

Rank is assigned by the pipeline and travels with its row, so re-sorting the grid changes the view and never the export order; an empty filter means unrestricted, so a cleared filter cannot blank the list by accident; and the export adds four traceability fields, so any downloaded row joins back to its evidence and to the exact pipeline inputs that produced it. There is no Top 500 or broader view — the published queue is fixed at 50 records, because a review capacity stated in advance is what makes the ordering meaningful.

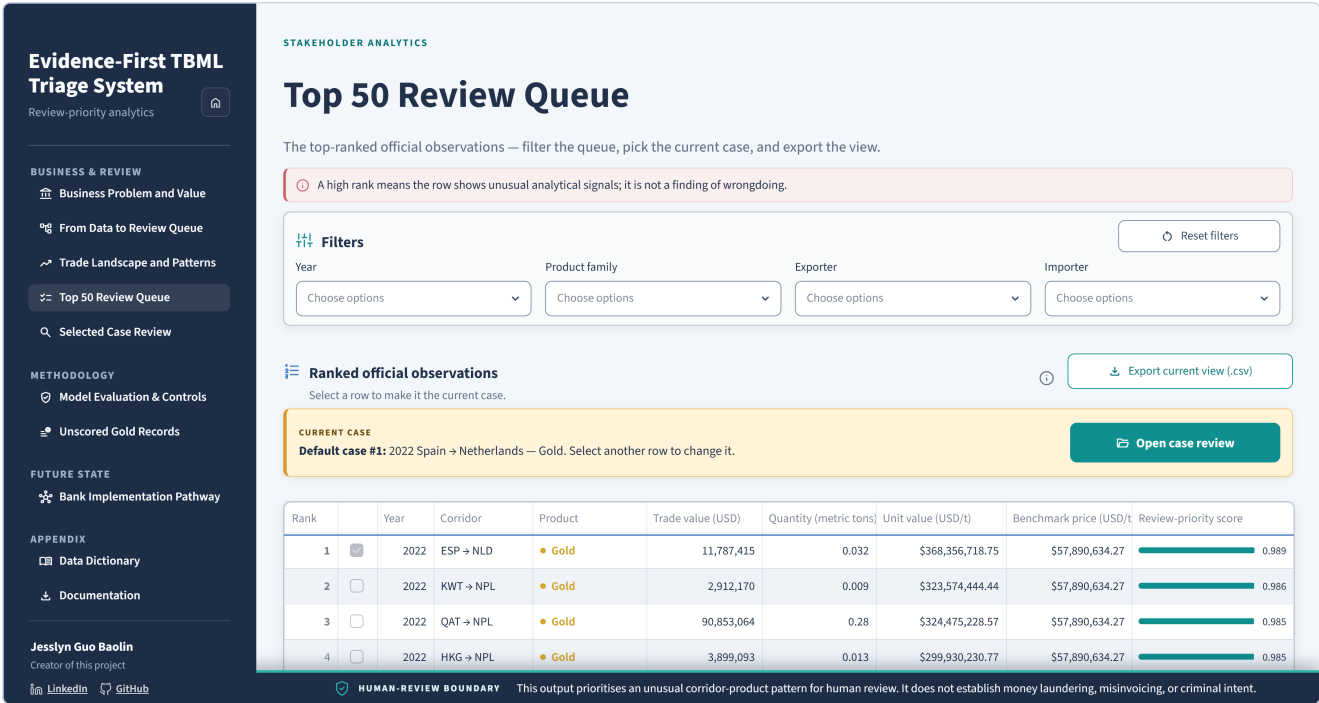


Figure 5. The Top 50 Review Queue as the deployed application renders it. Left: the navigation, grouped into Business & Review, Methodology, Future State and Appendix. Right: the interpretation caveat above the table, four filters scoped to the queue, the current-case banner with its hand-off into the case review, the export control, and the ranked grid with the review-priority score drawn as a proportional bar. The boundary ribbon is fixed across the foot of every page.

SECTION 05

The review queue and a selected case

The queue says where to start. The case view says why this record reached that position, what could otherwise explain it, and what a bank would still need to check.

The current queue contains 18 gold (36%) · 18 palm oil (36%) · 14 copper (28%), drawn from 34 distinct exporters and 33 distinct importers — so it is not concentrated in one trading relationship. The record below is the current rank-1 case, read live from the published queue rather than transcribed.

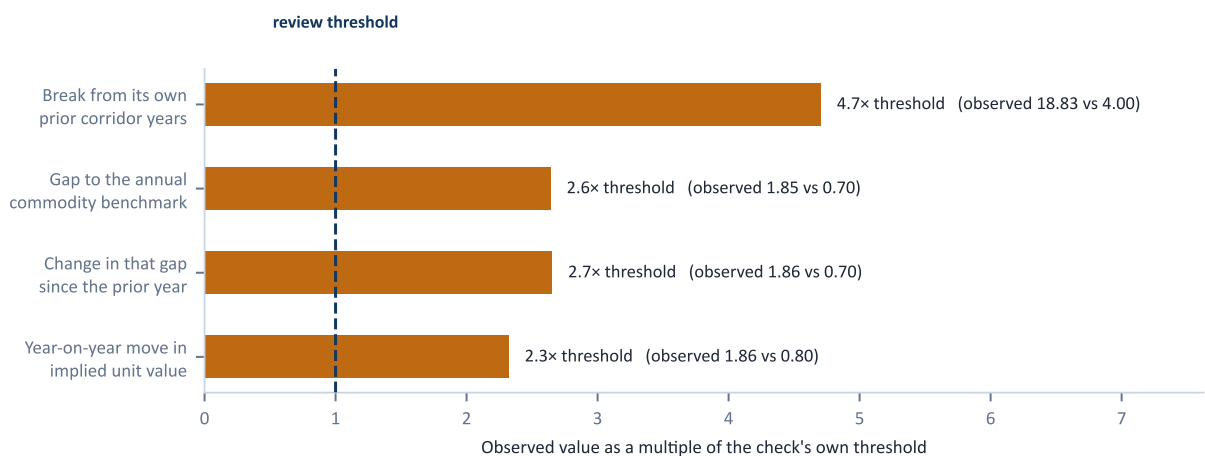
Current rank-1 record · 2022 Spain → Netherlands · non-monetary unwrought gold

Observation `obs_c1178b54e91326a9b0f5` , traceable to source row `baci_0805b6c6be2600ba0835c697` . Read live from the published queue.

REVIEW RANK #1 of 50 · score 0.9891	YEAR AND CORRIDOR 2022 · Spain → Netherlands ESP → NLD	PRODUCT Non-monetary unwrought gold HS6 710812
DECLARED TRADE VALUE \$11,787,415 annual public aggregate	DECLARED QUANTITY 0.032 metric tons usable, so the row can be scored	IMPLIED UNIT VALUE \$368,356,718.75 value ÷ quantity — not an invoice price
SAME-YEAR BENCHMARK \$57,890,634.27 World Bank annual commodity price	AGAINST THE BENCHMARK 6.36× benchmark 99.36% of 1,405 same-family peers that year	DATA QUALITY Usable with caveat 6 of 6 · valid extreme, checked and kept

What this record is not.

It is an unusual measurement in public annual data, not a finding about the exporter, the importer, either country, or any customer. Public statistics carry no purity, product form, freight, insurance, contract timing or invoice terms, and any of those could explain the figure. The rank says where to start, not what will be found.



Each check compares one measured quantity with the reference recorded beside it in the evidence table, and every check carries a caveat naming what could otherwise explain the value. A check passing its threshold is a reason to look, not a finding.

Figure 6. The four evidence checks behind the rank-1 record, each as a multiple of the threshold it was tested against. Plotting multiples rather than raw values keeps four quantities on different natural scales honestly comparable.

Reading the case honestly

Four independent comparisons agree on this record: it sits **6.36× the same-year World Bank benchmark**, breaks sharply from its corridor's prior years, shows a gap to the market that widened rather than tracking it, and moved year on year in implied unit value. Against 1,405 same-family corridors in 2022 it sits at the **99.36% percentile**. That agreement is what puts it first; it is not what would make it a finding.

Table 2. The same record, read as a review would read it.

What the public data shows	What could benignly explain it	What a bank would need to check
Implied unit value 6.36× the annual benchmark	Product form, purity or grade not visible in an HS6 aggregate	The commercial invoice and the goods description
A sharp break from the corridor's own prior years	A one-off contract, a changed counterparty, or a reporting revision	Trade-finance documents and the customer's expected activity
The gap to the market widened rather than tracking it	Freight, insurance and contract timing sit inside the declared value	Incoterms, shipment records and customs declarations
A very small declared quantity behind a large declared value	A rounding or unit convention in the reported quantity	The source declaration and the physical quantity recorded on it

Every row of this table is stated in the product itself: the middle column is the caveat carried by the evidence row, and the right-hand column is the follow-up recorded on the case's typology card.

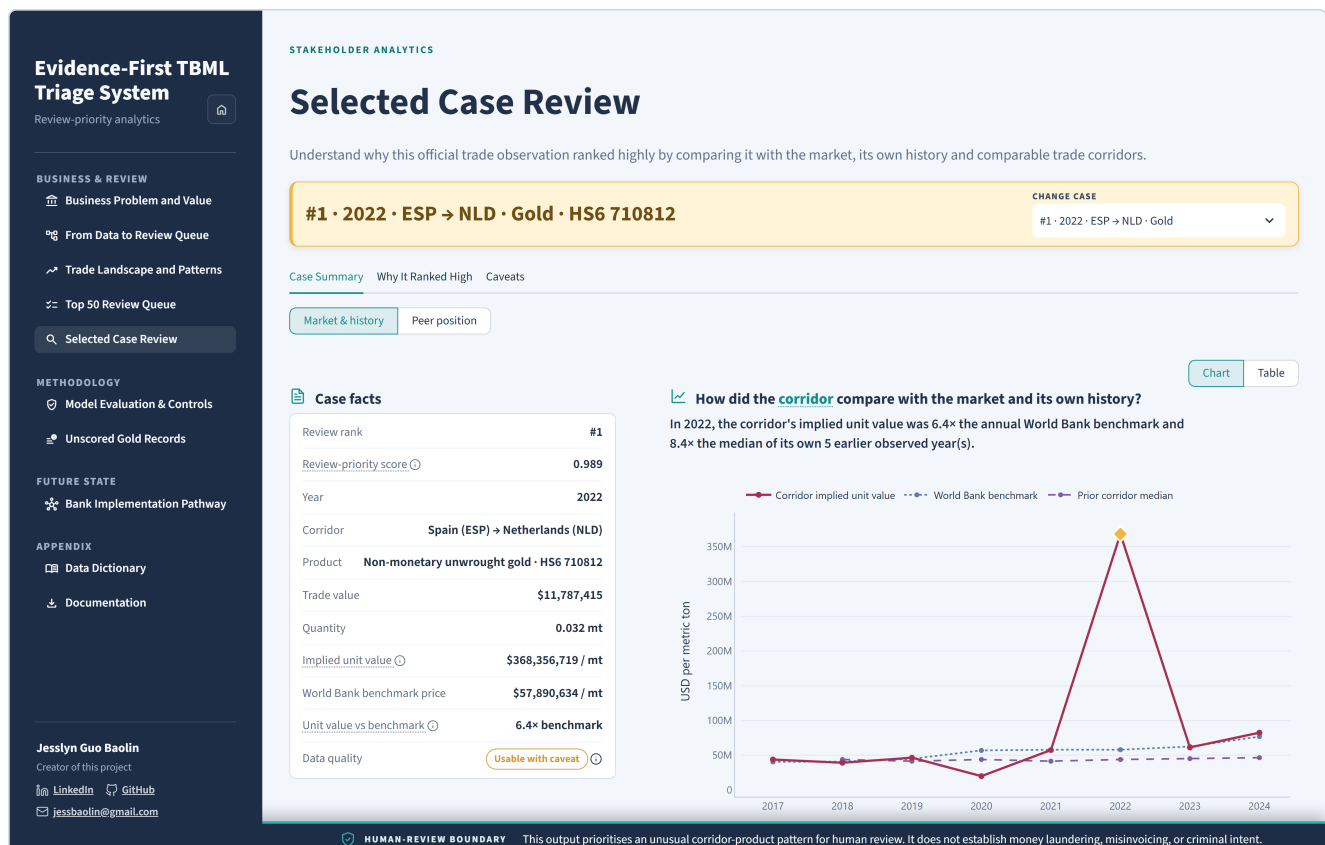


Figure 7. The Selected Case Review page in the deployed application. The amber strip fixes the case identity and offers the case selector; three tabs separate the summary, the ranking explanation and the caveats; the Case facts panel carries the observed values with their definition tooltips; and the comparison view plots the corridor's implied unit value against the annual benchmark and the median of its own prior observed years. Unobserved years stay blank rather than being interpolated. A runtime consistency gate guards the page: if any value recomputed on screen disagrees with the published queue, the feature table or the clean panel, the page shows the failure reason instead of the numbers — showing nothing is preferred to showing a figure that might be wrong.

SECTION 06

Stakeholder value and the implementation pathway

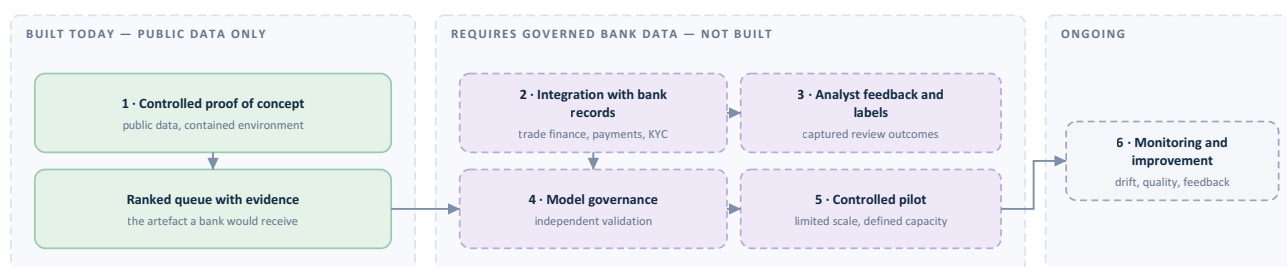
The value on offer is a defensible order of work and a consistent case pack — not a detection claim, and not a return-on-investment claim.

Table 3. What the project offers each stakeholder group in its current form.

Group	What it offers today
AFC analysts	A fixed-capacity queue with the reasons attached, so every case opens from stated facts and the same structure
Trade-finance teams	Independent corridor, product and period context, to sharpen a question before the document review
Relationship and portfolio teams	A portfolio view of unusual public trade behaviour, kept separate from customer risk rating
Data and analytics teams	A reproducible pipeline with time-safe features, a controlled evaluation design and hard negatives
Model-risk and governance	A method chosen on validation only, an over-alerting guard, and an auditable path from score to inputs
Senior management	A demonstration of evidence-first analytics: transparent, bounded, honest about what it cannot say

The benefit worth testing first is not efficiency. It is whether reviewers start from clearer facts and can document why a public signal was pursued or closed. Any claim about efficiency, detection or calibration would need a bank-controlled pilot and measured outcomes, which this project does not have and does not assert.

From a public-data signal to a governed bank review



Each phase adds controls the previous one did not need: access control, lineage, model-risk approval, audit logging, privacy and case-management integration.

None of phases 2 to 6 has been built. They describe what a bank would have to add, not what this project contains.

Figure 8. The route from what exists today to something a bank could operate. Only the first phase is built; everything drawn dashed requires records, controls and approvals that sit inside a bank and outside this project.

Each phase adds something the last did not need. **Integration** joins the queue to internal trade-finance, payment, KYC and screening records, so an aggregate pattern can become a decidable case. **Feedback** captures what reviewers found — evidence about the method, not ground truth, because dispositions carry selection bias. **Governance** adds independent validation, documented limitations, named approval and a monitoring plan; a **pilot** runs at limited scale against a defined capacity; and **monitoring** makes drift, quality and coverage an operating discipline.

CAUTION

Everything in phases 2 to 6 is design, not implementation. The project cannot move into production as it stands: it would need private bank records, access controls, model-risk approval, named analyst accountability, audit logging, privacy and security controls, and case-management integration. The repository contains none of these, and none is claimed.

SECTION 07

AI-assisted case preparation

The opportunity is not to automate judgement. It is to reduce case-preparation effort while preserving evidence traceability and human accountability.

CAUTION

Nothing in this section is built. The current project contains no AI or language-model component: no model is called, no key is configured, and nothing in the repository sends data anywhere. The analyst briefs that exist today are produced by a deterministic template that assembles already-computed evidence rows and refuses to emit a sentence the evidence does not support. That is a different thing from what follows.

Once a bank's own governed records sit beside the public-data signal, the expensive part of a review is no longer deciding what to look at. It is assembling the material: finding the related customer, trade-finance, shipment, customs, payment, screening and ownership records, comparing their details, and working out where they agree, where they conflict, and where something is simply missing.

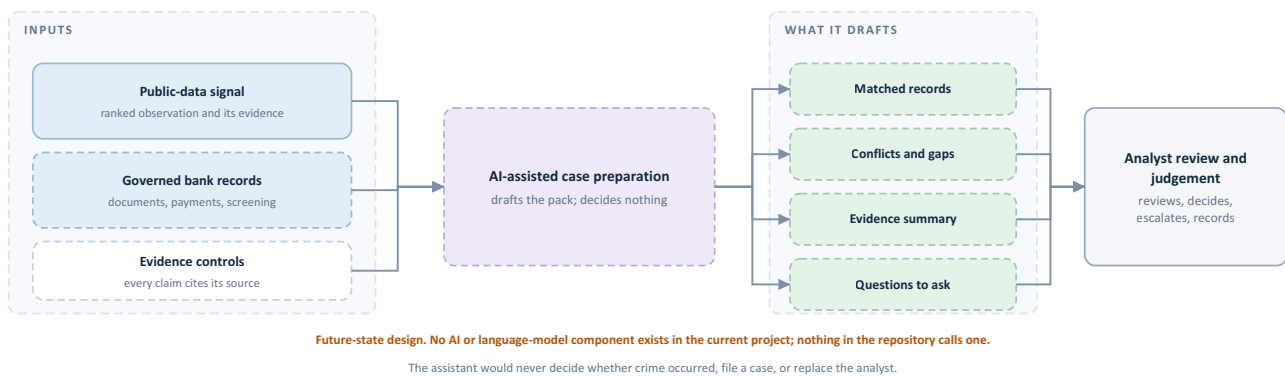


Figure 9. The future-state design. Governed records and the public-data signal enter under evidence controls; the assistant drafts the pack; every judgement, action and disposition stays with the analyst. Everything drawn dashed is unimplemented.

What it would look like in the product

The Bank Implementation Pathway page carries a labelled concept mockup of that design, so a stakeholder can see the shape of the thing rather than only read about it.

How it would behave inside the product Concept mockup

Selected Case Review · Rank 1 · ESP → NLD

Flagged case

ESP → NLD Gold · 2022

0.989 review-priority score

Trade value	\$11,787,415
Quantity	0.032 t
Unit value	\$368,356,718.75/t
Benchmark price	\$57,890,634.27/t

Evidence assistant
Connects records · never decides Powered by AI

Records compared

- ✓ Match Customer relationship fits the ESP → NLD gold route KYC/CDD ↗
- ✗ Conflict Invoice unit value sits far above the market benchmark Invoice #4471 ↗
- Gap No customs weight found for the declared 0.032 t Customs — none ↗
- ✓ Match Payment routed through the expected correspondent bank SWIFT MT103 ↗
- ⚠ Review Adverse-media item on a connected entity needs a look Adverse-media ↗
- Gap Ultimate owner of the buyer not yet confirmed Ownership — partial ↗

Ask the case

Any sanctions or adverse-media hits on the counterparties?

No sanctions match 1 item to review

No sanctions-list match on either counterparty. One adverse-media item on a connected entity — flagged for a separate human review.

Screened: sanctions & adverse-media service

Ask about this case... →

Answers cite their sources · the assistant does not decide the outcome.

Linked bank records Integrated

Open the source files behind this case.

KYC/CDD
Customer profile · 2 files

Trade-finance documents
Invoice, contract · 3 files

Shipment & customs
Bill of lading · 1 file

Payments & correspondent
SWIFT MT103 · 2 files

Sanctions & adverse media
Watchlist + media screen

Beneficial ownership
UBO structure · 2 files

Every fact links to its source. The decision stays with the analyst.

Figure 10. The evidence-assistant concept as the deployed application presents it, under a **Concept mockup** badge. Left: the flagged case and its public-data figures — the values this brief has already shown. Right: each retrieved bank record labelled **Match**, **Conflict**, **Gap** or **Review**, with its source named beside it, and a question-and-answer panel whose reply cites the service it screened. Below: the governed record sets that would be linked. Every figure in the panel is an illustrative placeholder, and no part of this exists in the current project.

What such a layer would and would not do

Table 4. The boundary that would have to hold for an assisted layer to be acceptable.

It would	It would never
Retrieve records related to the case from governed internal systems	Decide whether financial crime occurred
Show where those records match, conflict, or leave a gap	File, escalate or close a case automatically
Summarise the evidence already computed, with each point citing its source	Replace the analyst, or their accountability for the outcome
List the questions a reviewer should ask next	Generate a conclusion the underlying records do not support
Preserve the human-review boundary on every output	Bypass governance, access control or evidence requirements

The controls that make this acceptable are not novel: every factual statement traceable to a governed source record; no unsourced claim permitted to appear; the boundary statement carried on every output; and a named person responsible for the decision. Those are the same controls the deterministic brief layer already enforces today, which is what makes the extension a credible next step rather than a leap.

KEY POINT

An assisted layer would change how quickly an analyst can *see* the evidence. It would not change who decides, what counts as evidence, or what the project is allowed to conclude.

In closing

This project demonstrates something narrow and useful: that public, official-derived trade data can be turned into a defensible order of work, with the reasoning attached to each ranked record and the limits stated in the same breath as the result.

What it shows today is a complete, reproducible path from verified source files to a ranked queue with recomputable evidence, presented in a deployed application that never substitutes a number it cannot derive. What it does not show is any conclusion about conduct — and the discipline of keeping those two apart is the point, not a caveat on it.

Transparent evidence improves review quality for a simple reason: a reviewer who can see the comparison, the threshold and the caveat can disagree with them. A score alone offers nothing to disagree with, so it is either accepted or ignored. Inside a bank, this would need private records, controls, governance, monitoring and accountable reviewers before it could inform any decision. The public-data prototype is a starting point for that conversation, not a substitute for it.

COMPANION DOCUMENT

The **Comprehensive Technical Documentation and Implementation Guide** is the authoritative record for architecture, data lineage, formulas, feature definitions, model development and validation, reproduction commands and full data dictionaries. This brief is its executive companion and does not restate them.

Deployed application

<https://tbml-review-triage.streamlit.app/>

Source code

<https://github.com/jessybaolin/evidence-based-tbml-corridor-drift>

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